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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Full-stack AI Engineer with 3 years of industry experience building and shipping production **GenAI** products end to end at an asset manager: a **ReactJS** frontend, **Python FastAPI** microservices in Docker on **AWS**, **RAG** over PostgreSQL with **pgvector**, and tool-calling **agents** that run calculations and generate PDF reports. Started as one of two people on the first GenAI systems, worked directly with Compliance, Client Service and Investment users, released to real users and iterated on what broke; acted as de facto team lead for up to 4 interns and was twice recognised internally with Best Team Project (2024, 2025). Builds outside work too (Wingman, Pauti, Contact Reminder) and placed in three hackathons and datathons. Recently completed an MSc in Data Science at the University of Bristol, with a dissertation in partnership with AstraZeneca. Seeking a full-stack AI engineering role on a small team that ships quickly to enterprise customers.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Production GenAI Products: Agents, Retrieval & LLM Integration

- Designed and built **GenAI** applications end to end (**ReactJS** frontend, **FastAPI** backend, **RAG** logic, **pgvector** database and embeddings), starting as one of two people on the team, for Compliance, Client Service and Investment (**LlamaIndex** with **Azure OpenAI GPT-3.5 & Llama-3**); used department-based retrieval, with a separate **pgvector** database for each department so access stayed siloed.
- Built **agentic tool-calling** workflows where **AI agents** call **Python** functions as tools to run **SQL**-backed fund-comparison calculations and generate **PDF** comparison reports on user request.
- Built a **Python** extraction pipeline that turns over **300** unstructured daily emails from credit rating agencies into structured data, fuzzy-matches them to tracked companies and sends analysts a daily summary, **saving 60 manual analyst hours a month**; replaced **AWS Textract** with **Azure OpenAI** when a rating agency began sending emails as images, which also improved accuracy.
- Took the applications from prototype to production by releasing to real users and iterating on what broke; guardrailed responses through **system prompts**, versioned every prompt change in **Git** and had the **Compliance** department review each application before release. Allocated models by cost: open-source **Llama-3** self-hosted on a **GPU EC2** instance for the largest department (Client Service) and stronger **Azure OpenAI** outputs for decision makers.

Full-Stack Backend, Microservices & Secure Deployment

- Built the backend of internal **GenAI** applications as **Python FastAPI** microservices (uvicorn) in **Docker**, with a **ReactJS** frontend served by **Nginx** on a separate instance; designed the infrastructure on **AWS** using core services such as **EC2, S3, ECR** and **Lambda**, with separate frontend and backend repositories and containers and a one-click **Jenkins** and **GitLab CI/CD** pipeline for dev and prod.
- Implemented **RBAC** for the GenAI apps with a Python authentication function against the company **LDAP** server; restricted access to company private IPs using **AWS Security Groups**; deployed an **Nginx** reverse proxy,

one of the fixes that took the applications from repeatedly failing to passing the **VAPT** (vulnerability assessment and penetration testing); coordinated with the security team on port whitelisting and development safeguards.

Customer-Facing Delivery & Data

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), integrated with legacy data from analysts' Excel files; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.
- Developed **Power BI** dashboards for fortnightly credit report analysis, migrating financial analysts from Excel to Power BI and improving analyst efficiency by **10 hours a week**; then recreated the dashboards in the **ReactJS** frontend so the wider organisation could use the data without Power BI licensing costs.
- Worked directly with users in Client Service, Compliance and Investment to turn requirements into working tools that replace manual work; acted as **de facto team lead** for up to 4 interns in my final year; recognised with **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built; I designed the RAG-based AI assistant and worked on the scoring algorithm and data preprocessing for a tool, built as FastAPI microservices with a React and TypeScript frontend, that ranks cancer cell lines by target inclusion and exclusion genes and returns the evidence behind each pick; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app I built as a side project with 108 LLM personas across 12 archetypes (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). Behaviour steered by prompts; LLM output treated as untrusted: validated with Pydantic, repaired and cached by input hash; rate limits shaped the design (12 calls instead of 108, serialised calls, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git
- **Pauti:** offline-only expense-splitting Android app on a CRDT architecture. GitHub: https://github.com/Arishtanemi4/pauti.git **| Contact Reminder:** Android app that texts contacts on birthdays and anniversaries. GitHub: https://github.com/Arishtanemi4/contact-reminder.git
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. Architecture study comparing CNN/ViT/ResNet encoders and LSTM/attention decoders, benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **B.Tech Final Year Project:** Udacity self-driving car simulator. (A self-driving car trained using a **Convolution Neural Network (CNN)** model.)

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2, S3, ECR, Lambda, SageMaker), Azure OpenAI, LlamaIndex, Docker, Jenkins, GitLab, Git, Nginx, Linux, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic), SQL, R (dplyr, tidyverse, ggplot2), TypeScript and NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite